Computer Hardware Basics for Computer Systems Servicing

My reviewer for our CSS specialization: what each part does, how to handle it safely, and what to check when something goes wrong.

The main parts inside the system unit

- CPU (Central Processing Unit): carries out instructions and does the calculations, so people call it the brain of the computer. It sits in a motherboard socket and needs a cooler because it gets very hot.
- Motherboard: the main circuit board that connects everything. The CPU, RAM, storage, expansion cards and power supply all plug into it or connect through it.- RAM (Random Access Memory): short-term working memory. It holds the programs and data being used right now. RAM is volatile, which means everything in it disappears when the power is turned off.
- Storage: long-term memory where the operating system, apps and files stay even when the computer is off. It is non-volatile.
- Power supply unit (PSU): converts the AC electricity from the wall outlet into the lower DC voltages the parts need. Never open a PSU, because its capacitors can hold a dangerous charge even when unplugged.

RAM vs storage

A classmate said her laptop has "512 GB of RAM." That was really her storage.

- RAM is faster but smaller, usually measured in a few GB like 8 GB or 16 GB, and it is temporary.
- Storage is slower but much bigger, often hundreds of GB or a few TB, and it is permanent until you delete things.

HDD vs SSD

- HDD (Hard Disk Drive): stores data on spinning magnetic platters read by a moving arm. It is cheaper per GB and good for big storage, but slower, noisier and easily damaged if dropped while running.
- SSD (Solid State Drive): stores data on flash memory chips with no moving parts. It is much faster at booting and opening apps, silent, and tougher against bumps, but costs more per GB.

Input and output devices

- Input devices send data into the computer: keyboard, mouse, scanner, microphone, webcam.
- Output devices show or send results out of the computer: monitor, printer, speakers, headphones.
- Some devices do both, like a touchscreen or a headset with a mic.

Common ports

- USB (Universal Serial Bus): connects flash drives, mice, keyboards, printers and phones. USB Type-A is the flat rectangle, while USB Type-C is the small oval that can be plugged in either way.
- HDMI (High-Definition Multimedia Interface): carries both digital video and audio to a monitor, TV or projector through one cable.

- RJ-45: the connector at the end of an Ethernet cable, used for wired network or internet connections.

ESD safety

ESD means electrostatic discharge, the small static shock you feel after walking on a carpet. Even one too weak to feel can permanently damage chips.

- Wear an anti-static wrist strap clipped to a grounded surface, like the bare metal frame of the case, so your body's charge flows away safely.
- Keep parts in their anti-static bags until you install them.
- Hold cards and RAM by the edges, never by the gold contacts.
- Unplug the unit from the outlet before opening it.

Basic assembly order

A common order, adjusted depending on the case:

- Install the CPU into the motherboard socket, matching the triangle mark.
- Apply thermal paste if needed and attach the CPU cooler.
- Insert the RAM modules until the clips lock.
- Mount the motherboard into the case on its standoffs.
- Install the power supply.
- Install the storage drives.
- Add expansion cards such as a graphics card.
- Connect all power and data cables, then the front panel connectors.
- Do a final check, close the case, then power on.

POST and beep codes

When you press the power button, the firmware (BIOS or UEFI) first runs the POST, or Power-On Self-Test. It quickly checks that basic hardware like the CPU, RAM and graphics is working before the operating system loads.

If POST finds a problem, the computer may give a pattern of beeps called beep codes. On many systems, one short beep means POST passed. Repeated or long beeps point to a problem, often with RAM or video. The exact meaning depends on the BIOS maker, so always check the manual.

Basic troubleshooting steps

- Identify the problem: ask the user what happened and observe the symptoms.
- Establish a theory of the likely cause, starting with the simplest, like a loose cable.
- Test the theory to confirm the cause.
- Plan and carry out the fix.
- Verify that the whole system now works.
- Document what you found and what you did.

Quick check

- Q: Why is RAM called volatile? A: Its contents are lost when the power is turned off.

- Q: Give one advantage of an SSD over an HDD. A: It is faster and has no moving parts, so it handles bumps better.
- Q: What port is used for a wired Ethernet connection? A: RJ-45.
- Q: What does POST stand for? A: Power-On Self-Test.
- Q: Where do you clip the anti-static wrist strap? A: To a grounded surface, such as the bare metal frame of the computer case.